Nithish Krishnabharathi Gnani

Curriculum Vitae

nkgnani101@gmail.com
nithishkgnani.github.io
in nithish-k-gnani

Education

2013 - 2017 **Bachelor of Technology - Mechanical Engineering**, *National Institute of Technology Karnataka (NITK)*, Surathkal, Karnataka, India
NITK is a top 10 engineering institution under the National Institutional Ranking Framework (NIRF).

Work Experience

2019 - Present **Research Assistant -> Technical Associate**, *Indian Institute of Science (IISc)*, Bangalore
Department of Electronic Systems Engineering & Center for Networked Intelligence
IISc is India's top ranked university under NIRF

- Selected as a Technical Associate, Sponsored by Center for Networked Intelligence (CNI), a Cisco CSR initiative at Robert Bosch Center for Cyber-Physical Systems (RBCCPS).

2017 - 2018 **Team Leader**, *Hindustan Coca-Cola Beverages Private Limited*, Hyderabad
Maintenance and Total Productive Maintenance (TPM)

- Managed a team of 20+ associates to achieve consistent reliability of utility systems.

Research Experience and Projects

Details available at https://nithishkgnani.github.io/projects

Tactile internet and cyber-physical systems

Jun' 2023 - Present **Intercontinental teleoperation**, *IISc & TU Delft*, Under Prof. T V Prabhakar & Prof. R R Venkatesha Prasad

- Built the *remote domain* consisting of a robotic arm, depth camera using ROS and a dynamic drawing mechanism at IISc and interfaced it with the operator domain consisting of a haptic device at TU Deft.
- Trained two masters students from TU Delft on working with the remote domain
- *Novelty - Operator Intent:* A framework that extends Model Mediated Teleoperation to prioritise operator intent to tackle dynamic environments and high latency of up to 1 second.

Mar' 2022 - Present **Designing Tactile Cyber-Physical Systems**, *IISc*, Under Prof. Chandramani Singh & Prof. T V Prabhakar, Funders: Ministry of Electronics and Information Technology (MeitY), Govt. of India & Cisco CNI
Preprint

- Built a Tactile Cyber-Physical System (TCPS) testbed with a robotic arm and haptic device to demonstrate teleoperation for real-time interaction between humans and robots in real and virtual worlds for applications requiring ultra-reliable low latency communication (uRLLC).
- Programmed a Geomagic Touch haptic device with C/C++ to asynchronously get coordinates and velocity and send it to the robot as UDP packets. Programmed a UR3 robotic arm using ROS-Python to receive control packets, move to desired pose and send position feedback.
- *Novelty - EdgeP4:* Developed and implemented edge intelligence algorithms for teleoperation, *pose correction* and *tremor suppression* on P4-programmable network edge switch ports. Reduced control loop latency (<100 μs for *pose correction* task) and network load (99% reduction). Multiple algorithms can be hosted on the same edge switch which can transparently switch between the algorithms depending on the tasks.

Mar' 2021 - Present **Time Sensitive Networking (TSN) switch**, *IISc*, Under Prof. Chandramani Singh & Prof. T V Prabhakar, Funders: Ministry of Electronics and Information Technology (MeitY), Govt. of India & Cisco CNI
Preprint

- Building an IEEE 802.1 TSN capable ethernet switch by implementing in hardware, a. time synchronization (IEEE 802.1AS), b. Time Aware Shaper (IEEE 802.1Qbv), and c. packet duplication and elimination (IEEE. 802.1CB).
- *Novelty - μTAS:* Did a P4 - Micro C based system implementation of Time-Aware Shaper onto a programmable SmartNIC. Achieved a latency bound of 20 μs between two end hosts connected through two switches (built using SmartNICs).
- *Novelty:* Developed packet de-duplication algorithms for SmartNICs to efficiently eliminate duplicates for enhancing the reliability of Scheduled Traffic in Time-Sensitive Networks. For duplicating over two links with 10% packet losses in each, achieved perfect de-duplication for a single 2.5 Gbps stream. For 12 simultaneous streams of total 1 Gbps, obtained de-duplication efficiency of 99.88% with 99.83% packet delivery.

Jul' 2016 - Apr' 2017 **Yaw Control of a CubeSat Using Reaction Wheels**, *National Institute of Technology, Karnataka*, Bachelor's major project under Prof. Prasad Krishna
- Designed a suspended CubeSat model and achieved precise yaw control using reaction wheels using PID controller.
- Modeled it in Simulink and implemented Model Reference Adaptive Control (MRAC) using the MIT rule and achieved better settling time, overshoot, and steady-state error compared to PID control.

Wireless communication and video streaming

Mar' 2020 - June 2022 **5G AMMAZING**, *IISc*, Under Prof. K J Vinoy & Prof. T V Prabhakar
- An end to end 5G mmWave system for infotainment.
- *Novelty:* Data & control plane of a video stream split over 5G and regular Wi-Fi.
- First prize in 5G Hackathon 2020 by Dept. of Telecommunication, Government of India.

July 2020 - June 2021 **Wi-Fi modelling**, *IISc*, Under Prof. Neelesh Mehta & Prof. Chandramani Singh, Funder: Boeing
- Modelling & simulation of IEEE 802.11n & 802.11ac using MATLAB.
- Performance Metrics: throughput, packet error rate & range.
- Implementing Rate Control Algorithms in MATLAB.

Feb' 2020 - Jan 2021 **Video casting in dense environment**, *IISc*, Under Prof. T V Prabhakar, Funder: Boeing
- As 2.4 & 5 GHz Wi-Fi is congested in an aircraft cabin in which 400+ passengers cast media onto seat back displays in close proximity, using IEEE 802.11ad, 60GHz Wi-Fi (Wi-Gig) & 802.11ax (Wi-Fi 6E) was explored.
- MATLAB WLAN Toolbox was used to simulate the physical behavior of Wi-Fi transmissions in a dense environment.
- *Novelty:* Developed an algorithm that dynamically allocates different channels to 400 transmitters such that the packet error ratio is zero (PER = 0).

IoT data management, indoor localization and security

Nov 2019 - Jun' 2020, *Published at Wiley CCPE* **Data management of energy harvesting powered IoT sensors**, *IISc*, Under Prof. T V Prabhakar
- *Novelty:* Developed an *adaptive sampling* algorithm that adjusts the sampling rate and resolution of the temperature sensor based on the energy available in a vibration and heat harvesting sensor node.
- *Novelty: Regulating function* for data transmission that balances the amount of transmitted data and the reconstruction error at the aggregator based on the node's energy and the data characteristics, such as priority and rate of change.
- *Under publication:* Comparing the performance of Auto Regression, Long Short-Term Memory and dynamic programming based algorithms with adaptive sampling and regulating function.

Nov' 2019 - Jul' 2021, *Published at ACM DTRAP* **Acoustic Physically Unclonable Function (APUF) for sensor device security**, *IISc*, Under Prof. T V Prabhakar
- APUF is a novel technique to identify sensor devices and their positions by exploiting the physical property variations and acoustic fingerprinting.
- Applications: monitoring calibration-integrity and authentication in sensor-network deployments.
- Applied a K-Nearest Neighbors model, to evaluate the accuracy of the uniqueness signature and the position signature for different cases of accumulation and temperature.
- Achieved $> 99\%$ identification accuracy and scalability and displacement detection of 5 cm.
- *Novelty:* Unlike usual method of adding PUF circuitry while manufacturing the device, here PUF is extracted using microphones connected to any commercially available device with an ADC.

June 2019 - Feb' 2020 **Airplane IoT data management**, *IISc*, Under Prof. T V Prabhakar, Funder: Boeing
- Developed machine learning algorithms to classify anomalies and take corrective actions using real-time data generated from sensors from a MATLAB/Simulink model of an aircraft environmental control system.
- *Novelty:* Developed an algorithm that does linear interpolation between non-consecutive data points from time series data to adaptively store data based on a cost function which balances storage space savings and error in reconstruction of data. Achieved 96% savings in storage space.

Mar' 2019 - Jan' 2020 **Acoustics Based Localization**, *IISc*, Under Prof. Chandramani Singh & Prof. T V Prabhakar, Funder: Boeing
- Developed an application to locate multiple wireless edge devices with embedded microphones by using audio signals from speakers.
- Simulated a few localization algorithms using MATLAB. Implemented KNN fingerprinting based localization of the receivers (programmed Nordic nRF52940 based nodes using embedded C)
- Achieved localization accuracy of 98%.
- *Novelty:* Unlike the usual source localization, here, the receivers (mics) are localized when the locations of speakers are known.

Manufacturing plant

Nov' 2017 - Apr' 2018 **Total Productive Maintenance (TPM)**, *Hindustan Coca-Cola Beverages Pvt. Ltd.*
- Implemented TPM steps 4 and 5 in the bottling plant which led to the JIPM TPM Consistency Award.
- Performed root cause analysis (RCA) for equipment failures

Publications

Under review 2023	A realistic approach to realize long distance haptic bilateral teleoperation H.J.C. Kroep, **Nithish K Gnani**, RR Venkatesha Prasad, T V Prabhakar
arXiv Preprint 2023	*EdgeP4: A P4-Programmable Edge Intelligent Ethernet Switch for Tactile Cyber-Physical Systems* **Nithish K Gnani**, Joydeep P, Deepak C, Himanshu V, Soumya R, Kaushal M, T. V. Prabhakar, C Singh
arXiv Preprint 2023	*μTAS: Design and implementation of Time Aware Shaper on SmartNICs to achieve bounded latency* Joydeep Pal, Deepak Choudhary, **Nithish K Gnani**, Chandramani Singh, T. V. Prabhakar
ACM Journal 2021	*A novel approach for identification of sensor devices through Acoustic PUF, ACM DTRAP* Girish Vaidya, T.V.Prabhakar, **Nithish K Gnani**, Ryan Shah, Shishir Nagaraja
Wiley Journal 2021	*Judicious data management for sustaining an energy harvesting sensor node, Wiley CCPE* K Singh, P Shukla, Sachin S. M., **Nithish K Gnani**, Prabhakar T. V., Joy Kuri
Manuscript in preparation	Extension of Judicious data management for sustaining an energy harvesting sensor node **Nithish K Gnani**, T V Prabhakar

Teaching Experience, Demonstrations and Talks

2023	**Instructor**, *PG Level Advanced Certification Course in 5G Technologies with AI and Cloud*, Taught two batches how to design cyber-physical systems leveraging programmable networks, *NSE TalentSprint and IISc*
Demo	**Tactile Cyber-Physical Systems**
May 2023	ITU Workshop, *IISc*
Jul' 2022	6th annual symposium of cyber-physical systems (CyPhySS), *IISc*
Jul' 2022	Digital India Week, *Gandhinagar, Gujarat*
Talk	**Achieving bounded latency for time-sensitive applications**
Nov' 2023	Cisco-IISc Day, *IISc*
Sep' 2023	IBM-IISc Research Day, *IISc*

Certifications

1. Python Specialization – University of Michigan: Certificate
2. Deep Learning Specialization – DeepLearning.AI; 4/5 courses complete: 1, 2, 3, 4

Skills

Programming	Python, MATLAB, P4, C, C++, Embedded C, Shell
Hardware	Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino
Tools	Linux, Git, Docker, LaTeX
Others	Autodesk Fusion 360, 3D printing, video editing, implementing papers

Volunteering

Feb' 2019	Taught underprivileged girls at Yuwa School in rural Jharkhand, India. Coached students in tackling interviews. Prepared the team to attend the Laureus Awards 2019. Yuwa won in "Sports for Good" category.

Interests and Activities

3D Prints	Designing and 3D printing figurines and functional items for home and office.
Flying Club	Built and flew RC planes of different types in NITK. Conducted RC plane building workshops.
Swimming	Gold medal in 800 yards, bronze in 1500 yards and 100 yards relay in Spectrum 2023, IISc.

Referees

1. **Dr. T V Prabhakar**, *Chief Research Scientist*, Department of Electronic Systems Engineering, Indian Institute of Science
2. **Dr. Chandramani Singh**, *Associate Professor*, Department of Electronic Systems Engineering, Indian Institute of Science
3. **Dr. Prasad Krishna**, *Director*, National Institute of Technology (NIT) Karnataka and NIT Calicut and *Professor*, Department of Mechanical Engineering, NIT Karnataka